

Aflevering 1

Thursday, 6 February 2025 14.20

Problem 2.28 Use Eq. 2.29 to calculate the potential inside a uniformly charged solid sphere of radius R and total charge q . Compare your answer to Prob. 2.21.

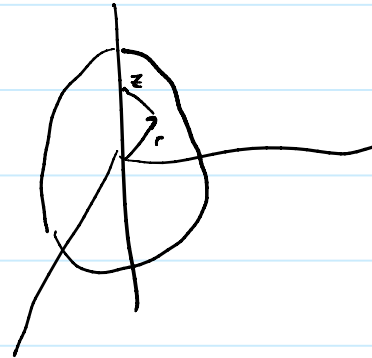
$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau'$$

$$r = \sqrt{r^2 + z^2 - 2rz \cos\theta'}$$

const. $\rightarrow$

$$V(\mathbf{r}) = \frac{\rho}{4\pi\epsilon_0} \int \frac{1}{r} d\tau$$

$$= \frac{\rho}{4\pi\epsilon_0} \int_0^R \int_0^{2\pi} \int_0^\pi \frac{r'^2 dr' \sin\theta d\theta d\phi}{\sqrt{r^2 + z^2 - 2rz \cos\theta}}$$



From example 2.8 we have $d\theta d\phi$:

$$= \frac{\rho}{4\pi\epsilon_0} \int_0^R \frac{2\pi r}{z} \left[\sqrt{(r+z)^2} - \sqrt{(r-z)^2} \right] dr$$

We solve the integral in parts:

$r > z$:

$$= \frac{\rho}{2\epsilon_0} \int_z^R \frac{r}{z} ((r+z) - (r-z)) dr$$

$$= \frac{\rho}{2\epsilon_0} \int_z^R 2r dr = \frac{\rho}{\epsilon_0} \left(\frac{R^2}{2} - \frac{z^2}{2} \right)$$

$r < z$:

$$= \frac{\rho}{2\epsilon_0} \int_0^z \frac{r}{z} (r+z-(z-r)) dr$$

$$= \frac{\rho}{2\epsilon_0} \int_0^z \frac{r^2}{z} dr = \frac{\rho}{3\epsilon_0} z^2$$

And then put them together:

$$V = \frac{\rho}{\epsilon_0} \left(\frac{R^2}{2} - \frac{z^2}{2} - \frac{z^2}{3} \right), \quad \boxed{\rho = \frac{3q}{4\pi\epsilon_0 R^3}}$$

$$= \frac{3q}{4\pi\epsilon_0 R^3} \left(\frac{R^2}{2} - \frac{z^2}{6} \right)$$

$$= \frac{q}{4\pi\epsilon_0 R} \left(\frac{3}{2} - \frac{z^2}{2R^2} \right)$$